

DL1

Fluid Flow through Fluidized Beds

Group #1
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Introduction

- The objective of this experiment was to determine the superficial minimum velocity of fluidization
- Conducting trials using sand and resin particles allowed for us to observe the differences in pressure drop and flowrate
- The minimum velocity was found using a combination of the Ergun equation and an empirical equation based on the Reynolds number of the particles

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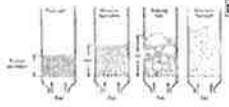
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Theory

Fluidized Bed is a layer of a fluidized solid, used in chemical processes.

Fluidized beds

- Fixed Bed
- Minimum Fluidization
- Bubbling Bed
- Pneumatic Transport



DL2

Gas Flow

- The change of air flow effects pressure on bed
- Beds are separated depending on pressure created by air flow
- Particle diameter and density can then be identified

$$\frac{\Delta P}{L} = \frac{150\mu U (1 - \epsilon)^2}{\epsilon^3 d_p^2} + \frac{1.75\rho U^2 (1 - \epsilon)}{\epsilon^3 d_p} \quad \text{DL3}$$

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Apparatus



Component #	Function
1 - Fluidized bed vessel	Plexiglass cylindrical tube that supports particles
2 - Porous metal disc	Provides a barrier between the air flow inlet and the particles in the bed
3 - Air flow valves	Controls flow rate of air into the vessel
4 - Rotameters	Reads the pressure of air flow into the system
5 - Manometer	Displays pressure drop across the bed

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DL1 97/100
Donglin Li, 10/29/2019

Slide 3

DL2 Good picture.
Donglin Li, 10/29/2019

DL3 What's the simplification of this equation used in this experiment, and how to simplify it?
Donglin Li, 10/29/2019

DL4 -1
Donglin Li, 10/29/2019

Slide 4

DL7 Some components are missing.
Donglin Li, 10/29/2019

DL8 -1
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Material & Supplies

- **Silica sand** - Quartz that over time, through the work of water and wind, has been broken down into tiny granules
- **Ion Exchange Resin** - Polymer that acts as a medium for ion exchange. It is an insoluble matrix normally in the form of small microbeads
- **Air** - Air tank is inserted to introduced air flow to the bottom of the vessel
- **Vacuum** - Used to clean the vessel after each experiment
- **Water** - Used to calculate void fraction
- **Scale** - To weigh beads when calculating particle diameter
- **Sieve trays** - To measure particle size

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Procedure

- Add the chosen particle to the vessel, pour from the top while standing on stairs. The Material should pack the vessel up to 6 inches for first run and 10 inches for second run
- Turn on Air, then slightly add air pressure to vessel flow in intervals of 10 until the recorded pressure drop on the manometer is zero.
- When the bed becomes fluidized record pressure drop and begin reducing the flow rate in increments until flow stops. Be sure to record pressure and pressure drop in system
- Repeat increase and decrease of air flow for 6in and 10in height using each particle. Total number of trials will be 4.

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Experimental Challenges

Experimental Challenges / Lessons Learned

- Adjust both rotameters so that the increments in flowrate stay consistent
- Save data files after each trial to prevent having to repeat trials
- There are two hoses for the two vacuum cleaners.
- When measuring void fraction use a larger graduated cylinder to make disposal easier ^{plus}
- Stack sieves with largest size on top and smallest on bottom
- Remove funnel at top of vessel when cleaning, not when adding material
- Tap vessel to level out the particles inside before starting trials

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Safety

- Wear safety glasses
- Wear gloves
- Read MSDS for resin and sand
- Emergency pressure release is located in the top right corner behind the apparatus
- Ladder is unstable, use care when climbing

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Slide 7

DL5 Could use bold style for key words at the beginning of each point to make it well-organized, like slide 5.

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